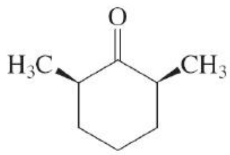
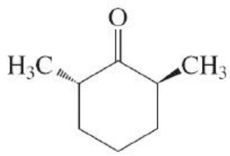
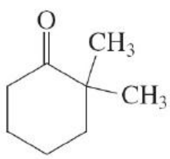
32. Underline the αα-carbons and circle the αα-hydrogens in each of the following structures.

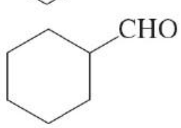
a. $CH_3CH_2\overset{\overset{O}{\|}}{C}CH_2CH_3$

b. $CH_3\overset{\overset{O}{\|}}{C}CH(CH_3)_2$

c. 

d. 

e. 

f. 

g. $(CH_3)_3C\overset{\overset{O}{\|}}{C}H$

h. $(CH_3)_3C\overset{\overset{O}{\|}}{C}H$

33. Write the structures of every enol and enolate ion that can arise from each of the carbonyl derivatives illustrated in Problem 32.

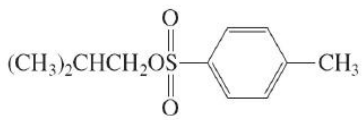
34. What product(s) would form if each carbonyl compound in Problem 32 were treated with **(a)** alkaline D2O;D_2O; **(b)** 1 equivalent of Br2Br_2 in acetic acid; **(c)** excess Cl2Cl_2 in aqueous base?

35. Describe the experimental conditions that would be best suited for the efficient synthesis of each of the following molecules from the

37. Give the product(s) that would be expected on reaction of 3-pentanone with 1 equivalent of LDA, followed by addition of 1 equivalent of

a. CH3CH2BrCH_3CH_2Br

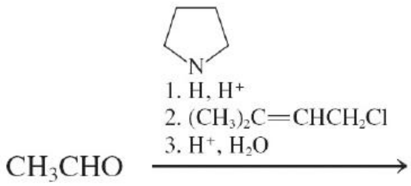
b. (CH3)2CHCl$(CH_3)_2CHCl$

c. 

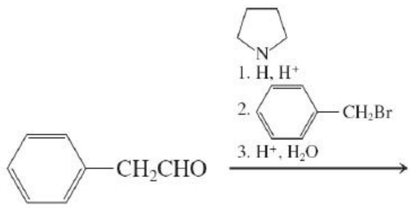
d. (CH3)3CCl$(CH_3)_3CCl$

38. Give the product(s) of the following reaction sequences.

a.



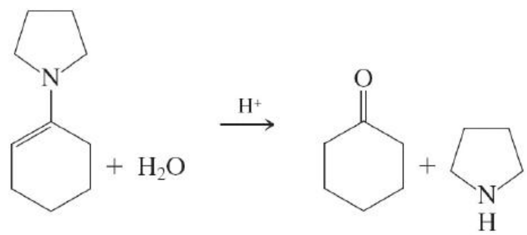
b.



39. The problem of double compared with single alkylation of ketones by iodomethane and base is mentioned in Section 18-4. Write a detailed mechanism showing how some double alkylation occurs even when only one equivalent each of the iodide and base is used. Suggest a reason why the enamine alkylation procedure solves this problem.

40. Would the use of an enamine instead of an enolate improve the likelihood of successful alkylation of a ketone by a secondary haloalkane?

41. Formulate a mechanism for the acid-catalyzed hydrolysis of the pyrrolidine enamine of cyclohexanone (shown below).

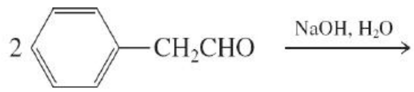


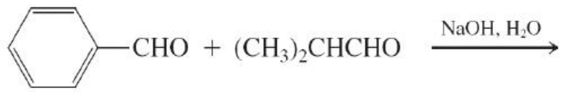
42. Write the structures of the aldol condensation products of **(a)** pentanal; **(b)** 3-methylbutanal; **(c)** cyclopentanone.

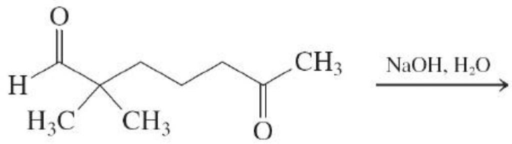
43. Write the structures of the expected major products of crossed aldol condensation at elevated temperature between excess benzaldehyde and **(a)** 1-phenylethanone (acetophenone—see Section 17-1 for structure); **(b)** acetone; **(c)** 2,2-dimethylcyclopentanone.

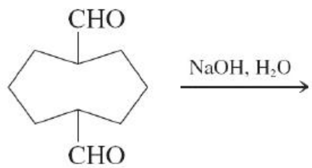
44. Formulate a detailed mechanism for the reaction that you wrote in Problem 43, part (c).

45. Give the likely products for each of the following aldol addition reactions.

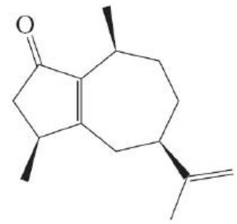
a. 

b. 

c. 

d. 

46. Rotundone (top of next column) is the natural product responsible for the peppery aroma in peppers, many herbs, and red wines (p. 885). What cyclic diketone will give rotundone upon intramolecular aldol condensation?

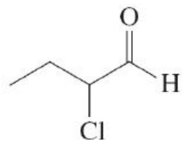


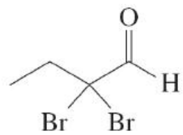
Rotundone

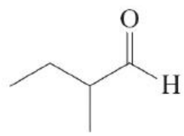
47. Write *all* possible products of the base-catalyzed crossed aldol reactions between each pair of reaction partners given below. (**Hint:** Multiple products are possible in every case; be sure to include thermodynamically unfavorable as well as favorable ones.)

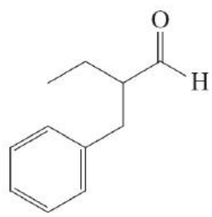
a. Butanal and acetaldehyde

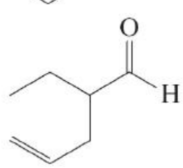
50. Reaction review. Without consulting the Reaction Road Maps on pp. 852–853, suggest reagents to convert butanal into each of the following products.

a. 

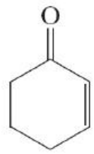
b. 

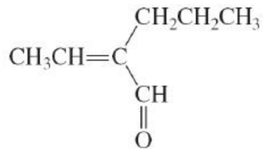
c. 

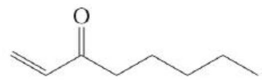
d. 

e. 

55. Write the expected major product of reaction of each of the carbonyl compounds (i)–(iii) with each of the reagents (a)–(h).

a. 

b. 

c. 

a. H2,Pd,CH3CH2OHH_2, Pd, CH_3CH_2OH

b. LiAlH4,(CH3CH2)2O$LiAlH_4$, $(CH_3CH_2)_2O$

c. Cl2,CCl4Cl_2, CCl_4

d. KCN,H+,H2OKCN, H^+, H_2O

e. CH3Li,(CH3CH2)2OCH_3Li, $(CH_3CH_2)_2O$

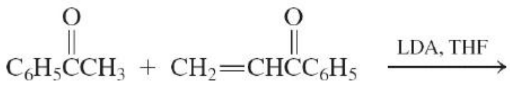
f. (CH3CH2CH2CH2)2CuLi,$(CH_3CH_2CH_2CH_2)_2CuLi$, THF

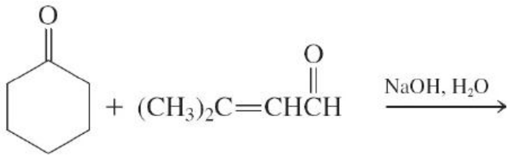
g. $NH_2NH\underset{\underset{\displaystyle O}{\|}}{C}NH_2$, CH_3CH_2OH

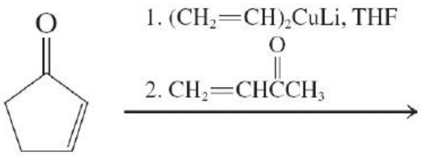
h. (CH3CH2CH2CH2)2CuLi,$(CH_3CH_2CH_2CH_2)_2CuLi$, followed by treatment with CH2═CHCH2ClCH_2â $CHCH_2Cl$ in THF

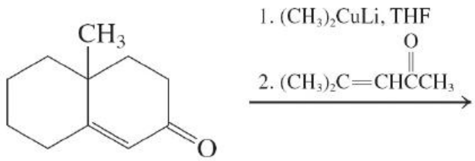
56. In each of the following retrosynthetic disconnections (), show a

57. Write the products of each of the following room temperature reactions after aqueous work-up.

a. 

b. 

c. 

d. 

e. Write the results that you expect from base treatment of the products of reactions (c) and (d) in boiling solvent.